

Problem Statement:

The Global Superstore dataset consists of a large amount of transactional sales information, including details about customers, products, regions, dates, sales, quantities, discounts, and profits. Since descriptive details such as customer and product information are repeated across many transactions, the raw dataset can become complex and difficult to manage for analysis and reporting. The objective is to convert this raw transactional data into a structured dimensional data model using Microsoft Power BI. The transactional and measurable data will be stored in a central Fact_Sales table, while descriptive attributes will be separated into dimension tables such as Dim_Customer, Dim_Product, Dim_Region, and Dim_Date. These tables will be linked using suitable one-to-many relationships, creating a Star Schema that makes the data easier to understand and analyze.

The developed model will allow business performance to be examined from different perspectives, including customers, products, regions, and time periods. Furthermore, DAX measures such as Total Sales, Total Profit, Profit Margin, and Average Order Value can be created to calculate important business metrics. These measures can be presented through Power BI charts, cards, slicers, and interactive dashboards, enabling users to identify trends, compare performance, and make informed business decisions.

Solution:

- Develop a Fact_Sales table to store individual sales transactions along with important numerical values and measures.
- Create separate Dim_Customer, Dim_Product, Dim_Region, and Dim_Date tables to organize descriptive information.
- Use Power Query to clean the dataset by removing irrelevant columns and eliminating duplicate records from the dimension tables.
- Define appropriate one-to-many relationships between the dimension tables and the central Fact_Sales table to build a Star Schema.
- Create essential DAX measures, including Total Sales, Total Profit, Profit Margin, and Average Order Value, for business performance analysis.
- Design interactive Power BI reports using charts, cards, slicers, and other visual elements to analyze sales performance from different business perspectives.

12.64M

Total Sales

1.47M

Total Profit

0.12

Profit Margin

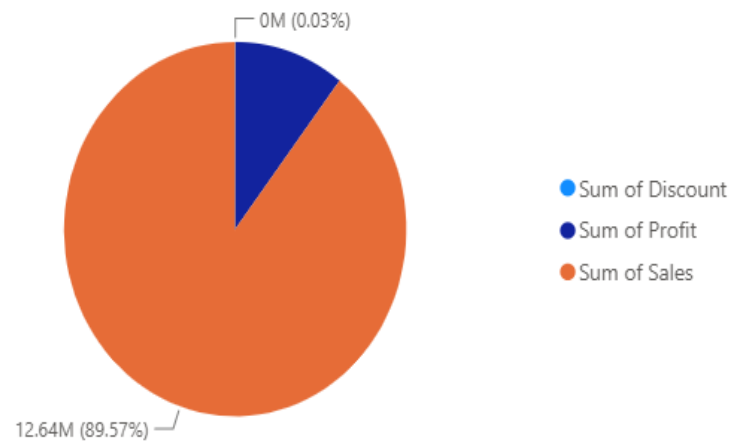
246.49

Average Order Value

178K

Total Quantity

Sum of Discount, Sum of Profit and Sum of Sales



Sum of Discount, Sum of Profit, Sum of Sales and Sum of Quantity

